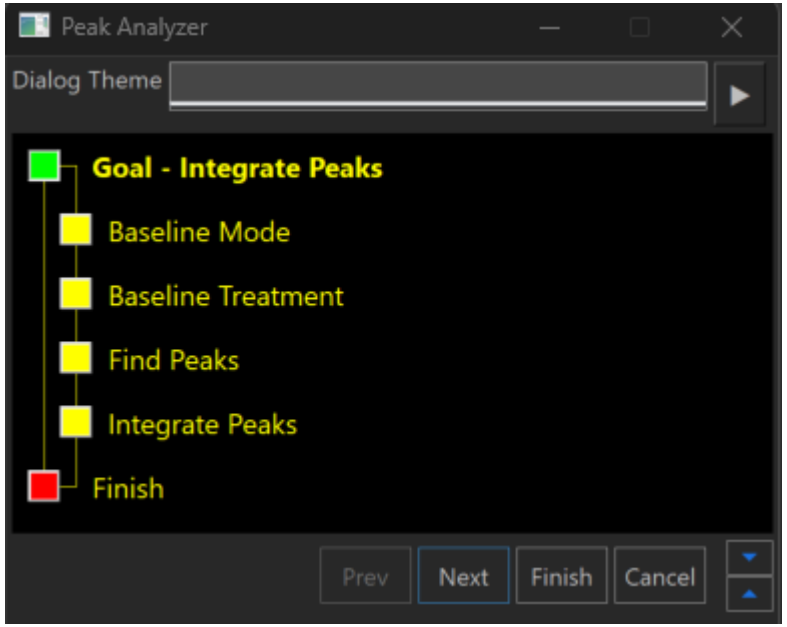
PL Data process (Origin+Python)

- Plot original data
- Find background & Fit main peaks

  (Bg, phonon side band: Gaussian; Main peak: Lorentz)
- Minus then fit side band peaks
- Draw with Python




```
tabnine: test | explain | document | ask
> def gaussian(x, amplitude, mean, std_dev):···

tabnine: test | explain | document | ask
> def lorentz(x, amplitude, mean, gamma):···

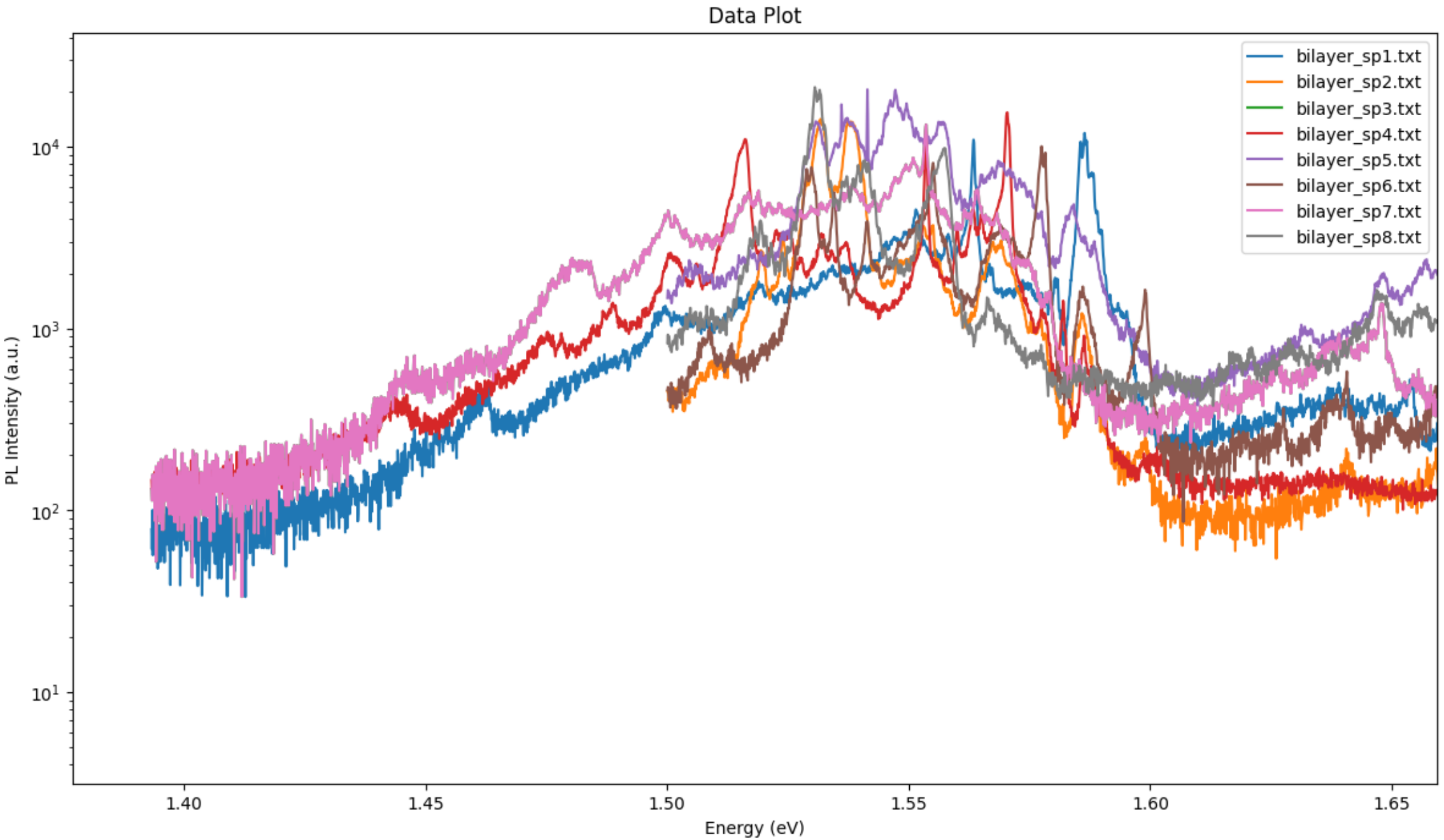
tabnine: test | explain | document | ask
> def voigt(x, amp,x0, gamma, sigma):···

tabnine: test | explain | document | ask
> def plot_lines(x_values, line_type, amplitude, mean, gamma, sig

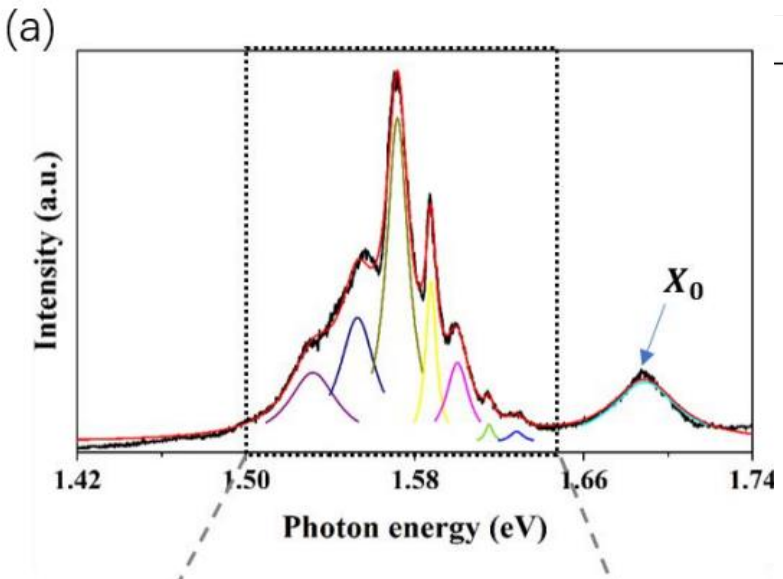
# Example usage
file = 'bilayer_sp6.txt'
data = np.loadtxt(file, delimiter=',')
x_values = data[:, 0]
y_values = data[:, 1]
y_corrected=y_values-gaussian(x_values,1500 ,1.55308, 0.046)
plt.plot(x_values,y_corrected , label='Data')
np.savetxt('no bd_sp6.txt', np.column_stack((x_values, y_correc
plt.plot(x_values, y_values,'r')

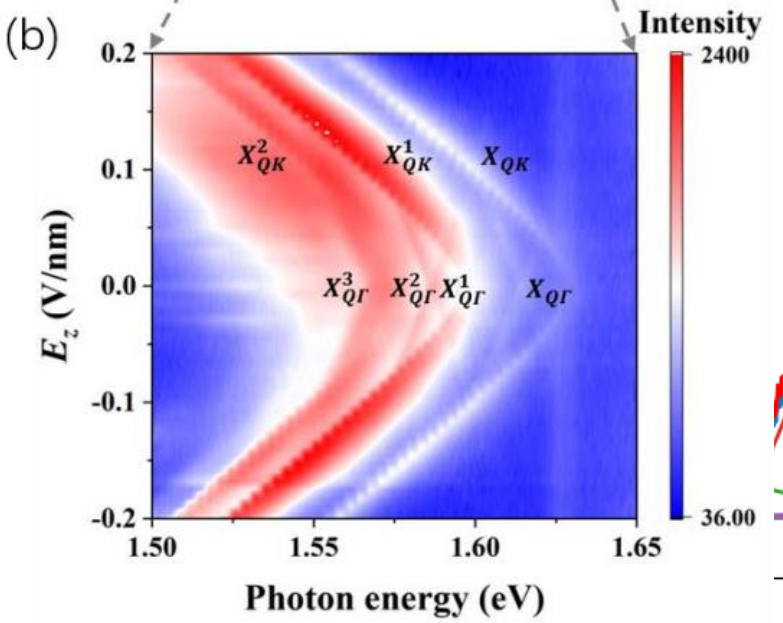
plot_lines(x_values, 'lorentz', 2588, 1.55272, 0.00476, 0, 0)
plot_lines(x_values, 'lorentz', 4485, 1.55502, 0.00113, 0, 0)
plot_lines(x_values, 'gaussian', 1203, 1.58636, 0.00313,0, 0)
plot_lines(x_values, 'gaussian', 1172, 1.59851, 0.00309,0, 0)
# main peak
#plot_lines(x_values, 'gaussian', 2188, 1.57048,0.01131,0,0)
plot_lines(x_values, 'lorentz', 8086, 1.57741,0.00184,0,0)
plot_lines(x_values, 'lorentz', 4472, 1.57830,0.00039,0,0)
```

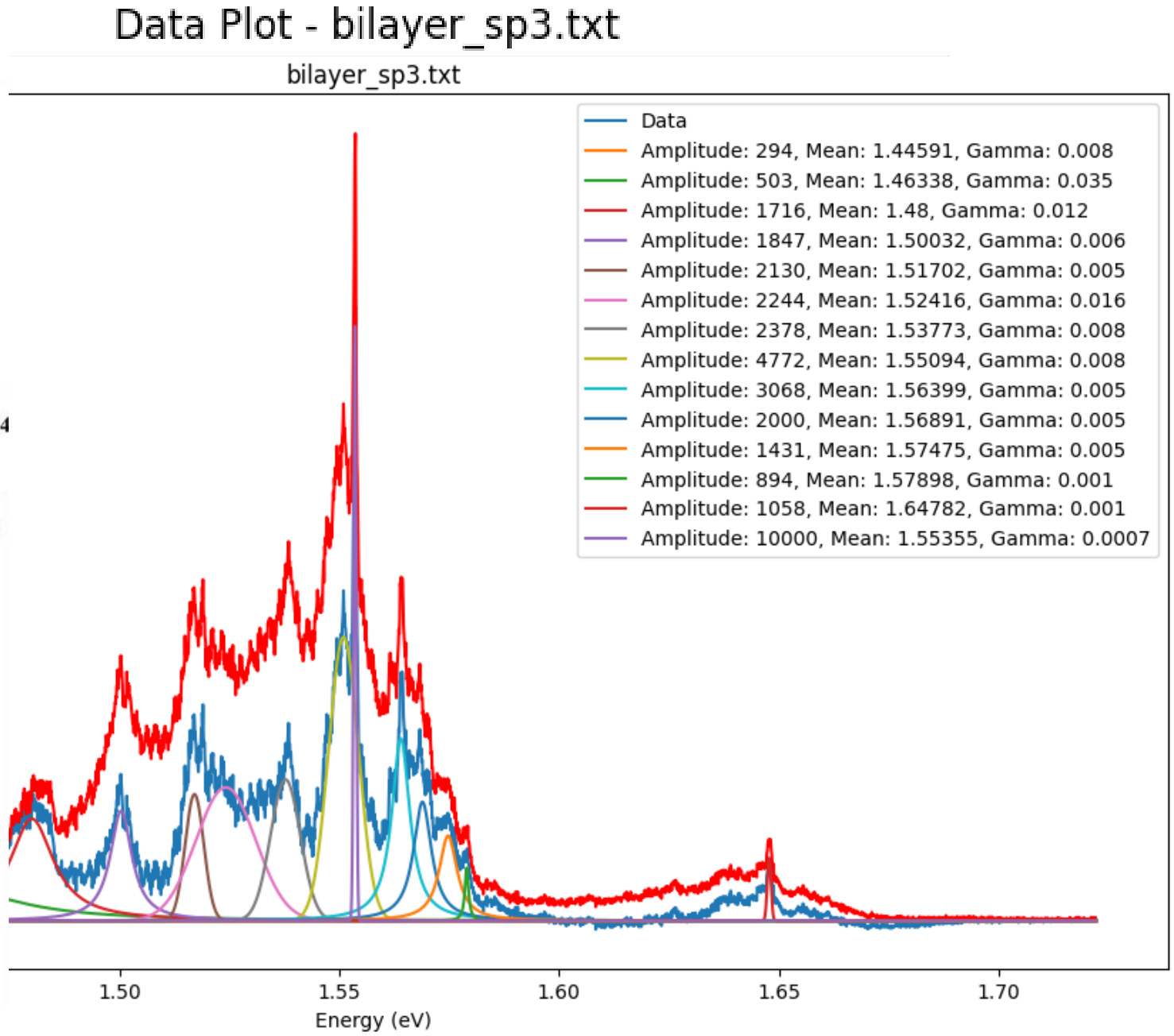
All spots


Data Plot
bilayer_sp1.txt
bilayer_sp2.txt
bilayer_sp3.txt
bilayer_sp4.txt
bilayer_sp5.txt
bilayer_sp6.txt
bilayer_sp7.txt
bilayer_sp8.txt
PL Intensity (a.u.)
10^4
10^3
10^2
10^1
1.40
1.45
1.50
1.55
1.60
1.65
Energy (eV)

Spot3

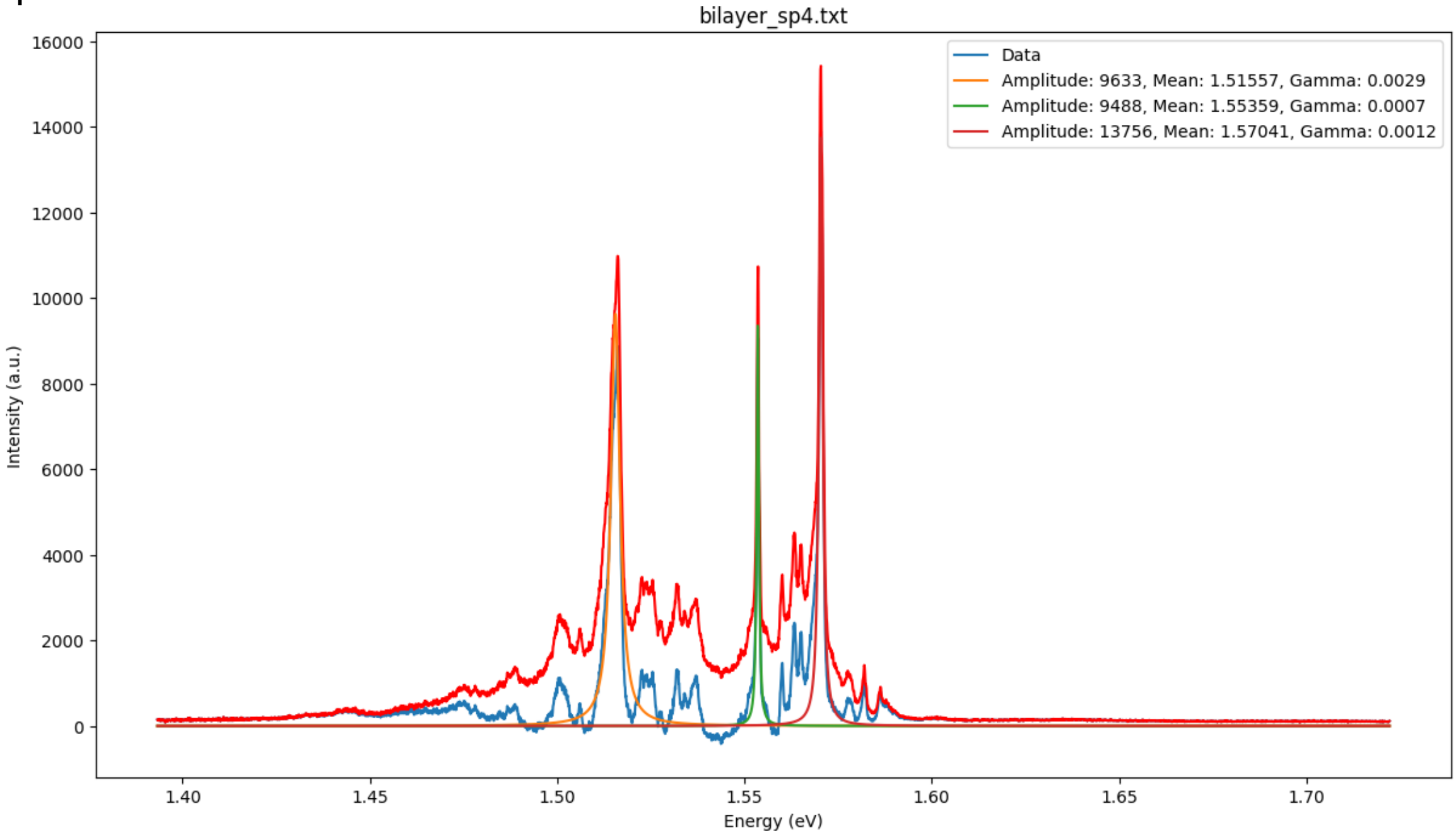




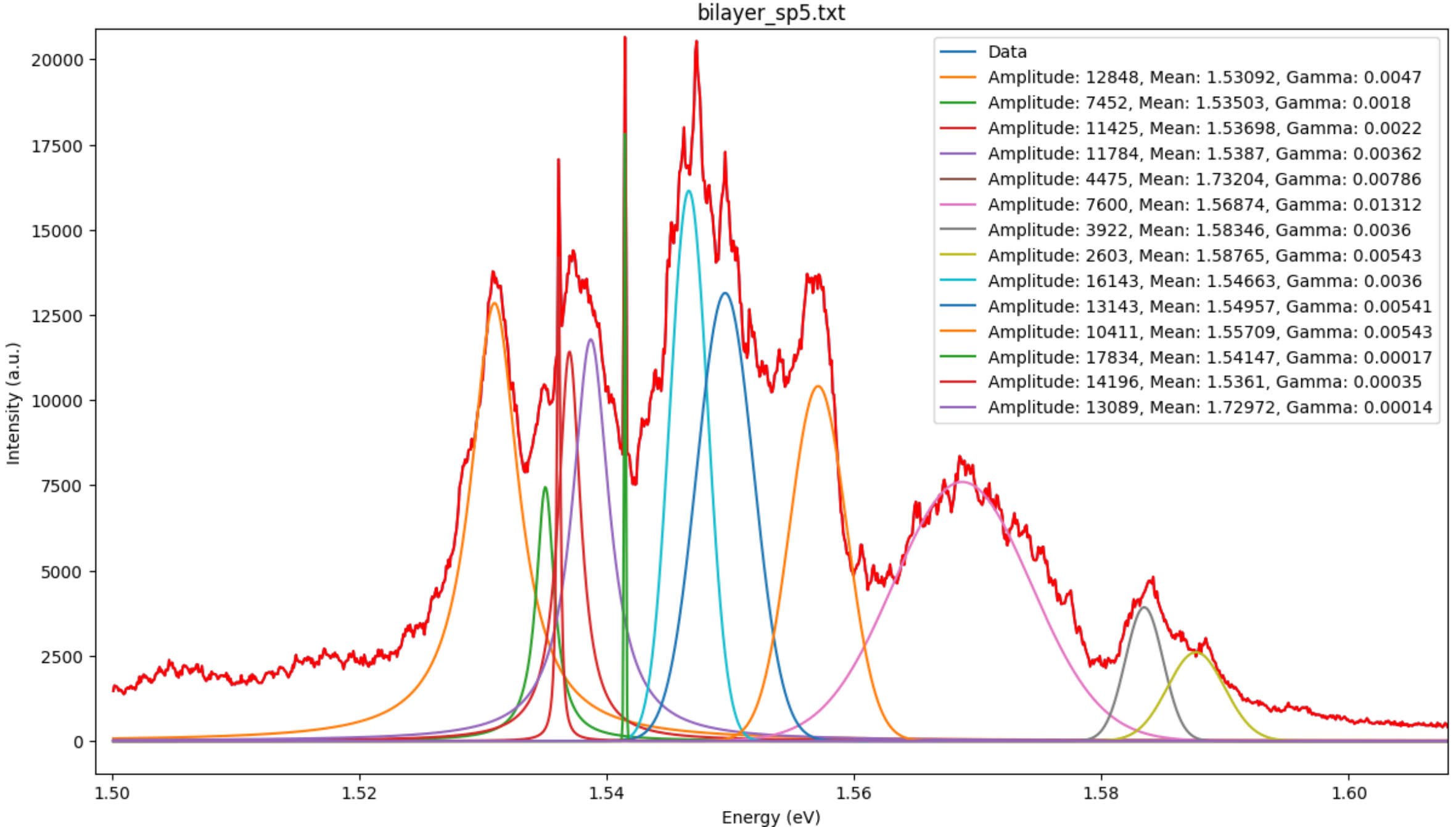


PhysRevB.105.L041409

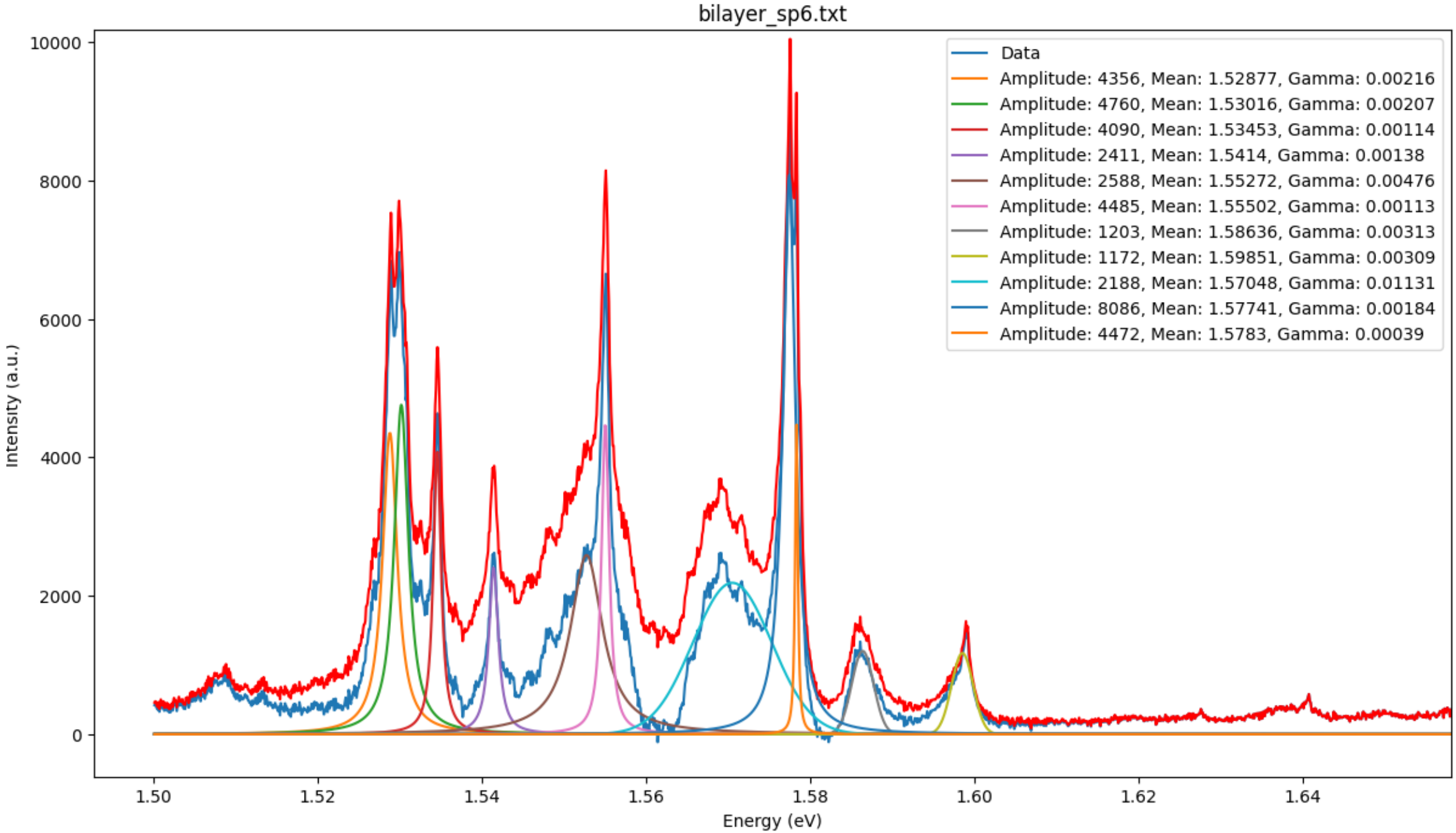
Spot4


bilayer_sp4.txt
Data
Amplitude: 9633, Mean: 1.51557, Gamma: 0.0029
Amplitude: 9488, Mean: 1.55359, Gamma: 0.0007
Amplitude: 13756, Mean: 1.57041, Gamma: 0.0012
Intensity (a.u.)
16000
14000
12000
10000
8000
6000
4000
2000
0
1.40
1.45
1.50
1.55
1.60
1.65
1.70
Energy (eV)

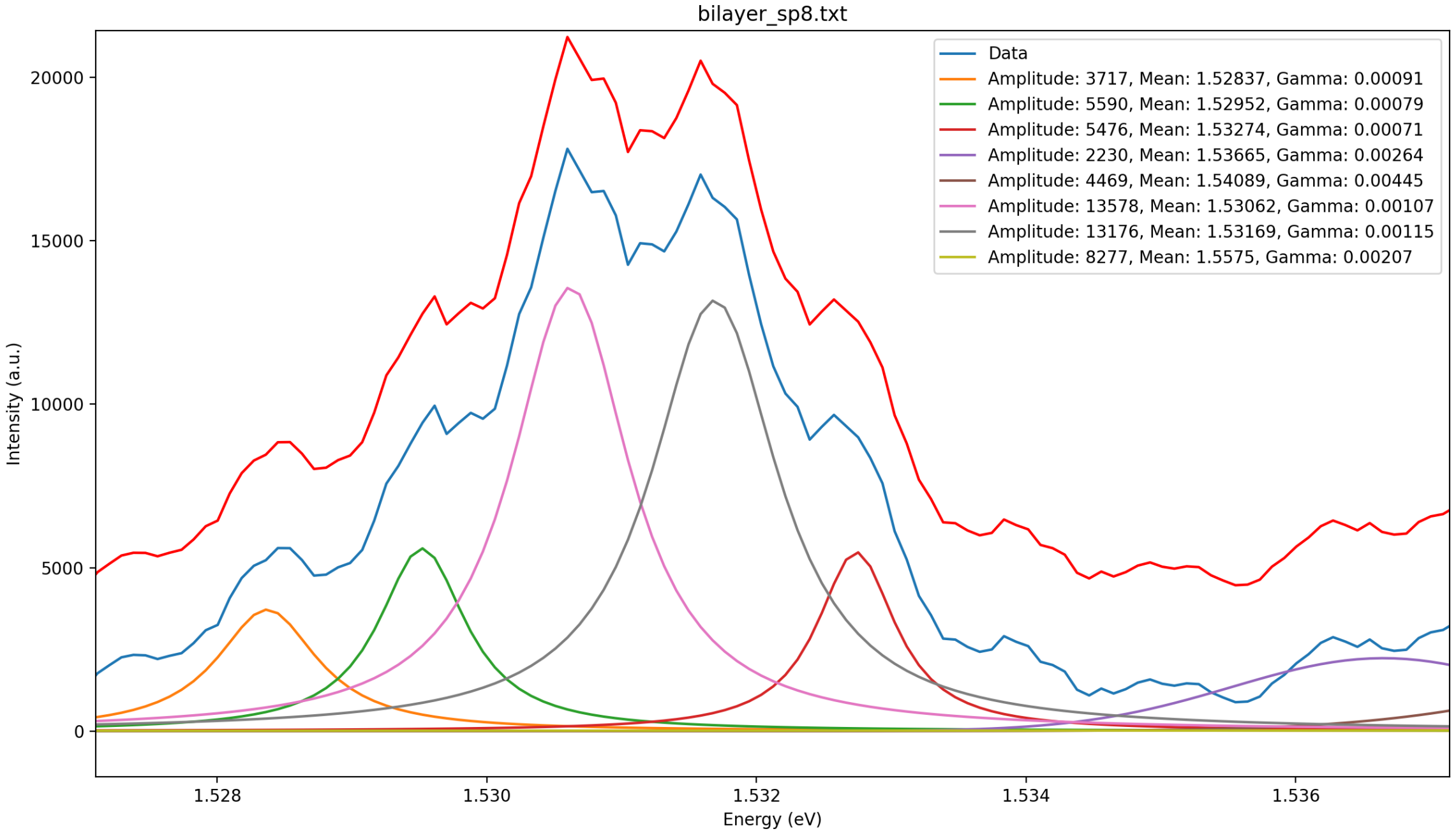
Spot5


bilayer_sp5.txt
Data
Amplitude: 12848, Mean: 1.53092, Gamma: 0.0047
Amplitude: 7452, Mean: 1.53503, Gamma: 0.0018
Amplitude: 11425, Mean: 1.53698, Gamma: 0.0022
Amplitude: 11784, Mean: 1.5387, Gamma: 0.00362
Amplitude: 4475, Mean: 1.73204, Gamma: 0.00786
Amplitude: 7600, Mean: 1.56874, Gamma: 0.01312
Amplitude: 3922, Mean: 1.58346, Gamma: 0.0036
Amplitude: 2603, Mean: 1.58765, Gamma: 0.00543
Amplitude: 16143, Mean: 1.54663, Gamma: 0.0036
Amplitude: 13143, Mean: 1.54957, Gamma: 0.00541
Amplitude: 10411, Mean: 1.55709, Gamma: 0.00543
Amplitude: 17834, Mean: 1.54147, Gamma: 0.00017
Amplitude: 14196, Mean: 1.5361, Gamma: 0.00035
Amplitude: 13089, Mean: 1.72972, Gamma: 0.00014
Intensity (a.u.)
0
2500
5000
7500
10000
12500
15000
17500
20000
1.50
1.52
1.54
1.56
1.58
1.60
Energy (eV)

Spot6


bilayer_sp6.txt
Data
Amplitude: 4356, Mean: 1.52877, Gamma: 0.00216
Amplitude: 4760, Mean: 1.53016, Gamma: 0.00207
Amplitude: 4090, Mean: 1.53453, Gamma: 0.00114
Amplitude: 2411, Mean: 1.5414, Gamma: 0.00138
Amplitude: 2588, Mean: 1.55272, Gamma: 0.00476
Amplitude: 4485, Mean: 1.55502, Gamma: 0.00113
Amplitude: 1203, Mean: 1.58636, Gamma: 0.00313
Amplitude: 1172, Mean: 1.59851, Gamma: 0.00309
Amplitude: 2188, Mean: 1.57048, Gamma: 0.01131
Amplitude: 8086, Mean: 1.57741, Gamma: 0.00184
Amplitude: 4472, Mean: 1.5783, Gamma: 0.00039
Intensity (a.u.)
10000
8000
6000
4000
2000
0
1.50
1.52
1.54
1.56
1.58
1.60
1.62
1.64
Energy (eV)

Spot8


bilayer_sp8.txt
Data
Amplitude: 3717, Mean: 1.52837, Gamma: 0.00091
Amplitude: 5590, Mean: 1.52952, Gamma: 0.00079
Amplitude: 5476, Mean: 1.53274, Gamma: 0.00071
Amplitude: 2230, Mean: 1.53665, Gamma: 0.00264
Amplitude: 4469, Mean: 1.54089, Gamma: 0.00445
Amplitude: 13578, Mean: 1.53062, Gamma: 0.00107
Amplitude: 13176, Mean: 1.53169, Gamma: 0.00115
Amplitude: 8277, Mean: 1.5575, Gamma: 0.00207
Intensity (a.u.)
20000
15000
10000
5000
0
1.528
1.530
1.532
1.534
1.536
Energy (eV)

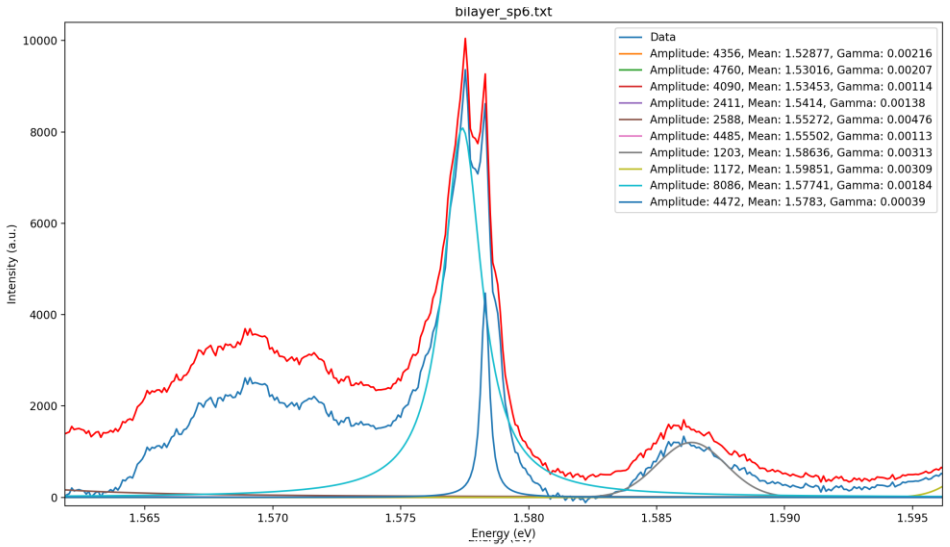
Analyze

Spot	FWHM/meV	Gaussian/meV	Lorentz/meV	Energy dif/meV	Lifetime? /ps
Spot3	8.15	8.02	0.70	2.61	0.94
Spot5	6.27	5.41	3.60(0.14)	2.94	4.70
Spot6	5.31	4.76	1.13(0.39)	2.30	1.71
Spot8	4.35	3.80	2.07(0.71)	2.54	0.93

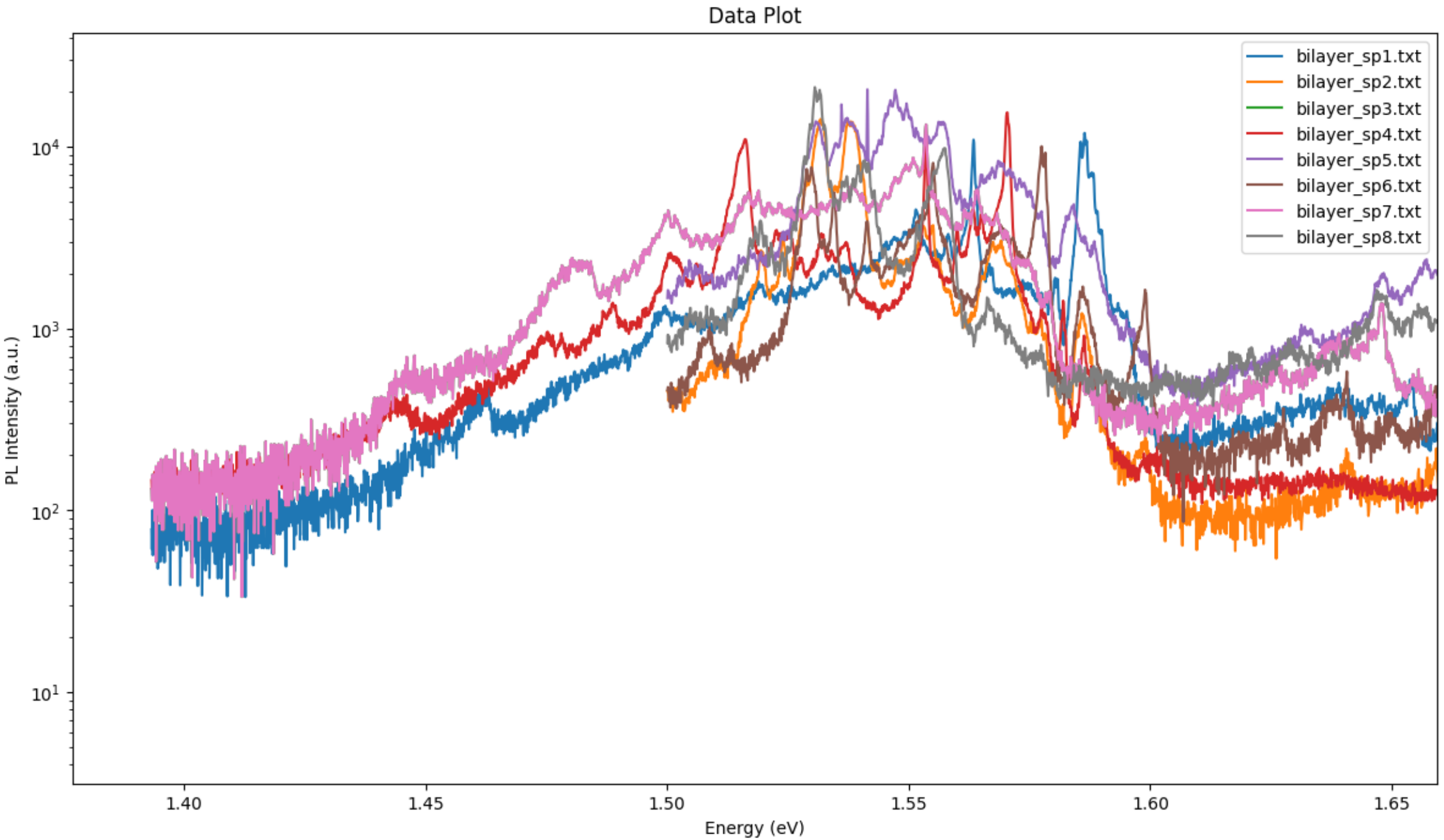
FWHM: Total emission linewidth

Lorentz: ZPL linewidth Gaussian: PSB linewidth
Energy difference: ZPL and PSB (Fig1)

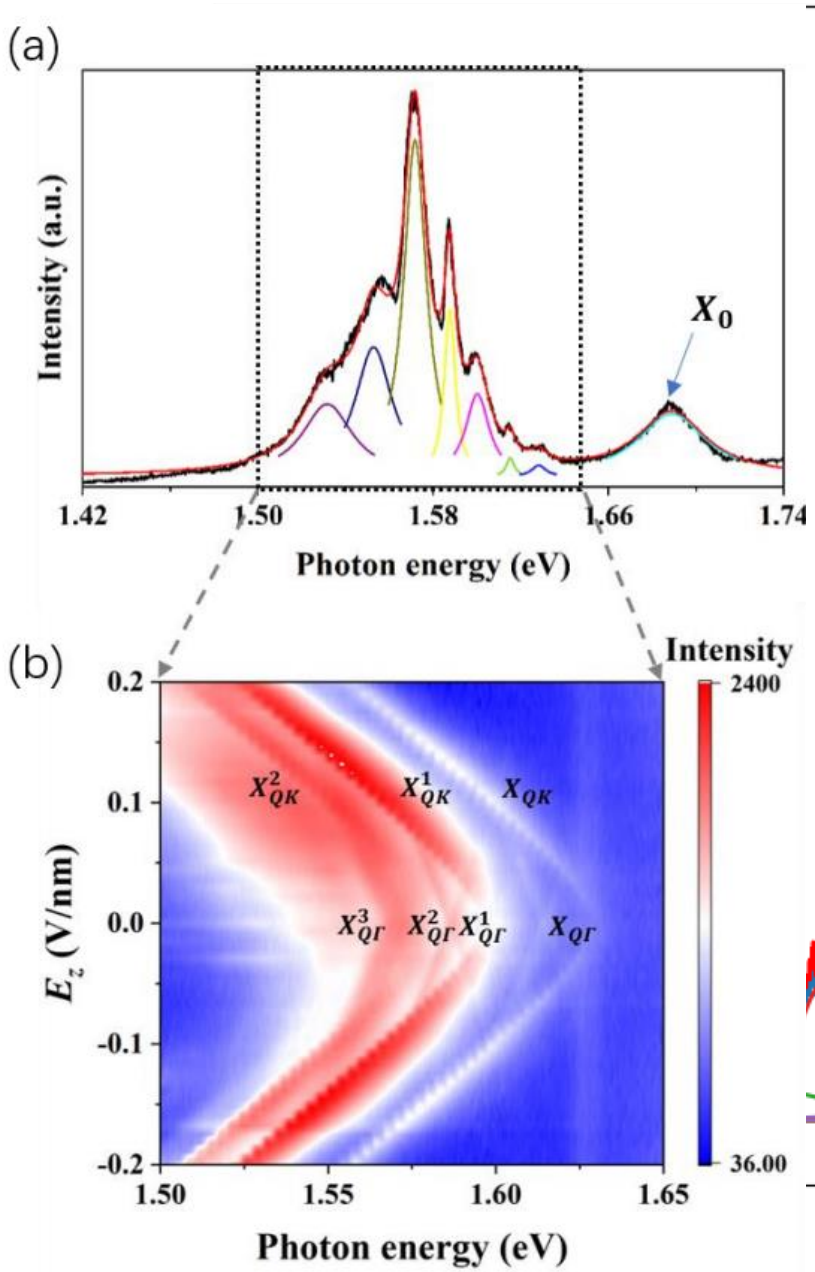
Energy difference between main peaks?: 0.9meV (Fig2)

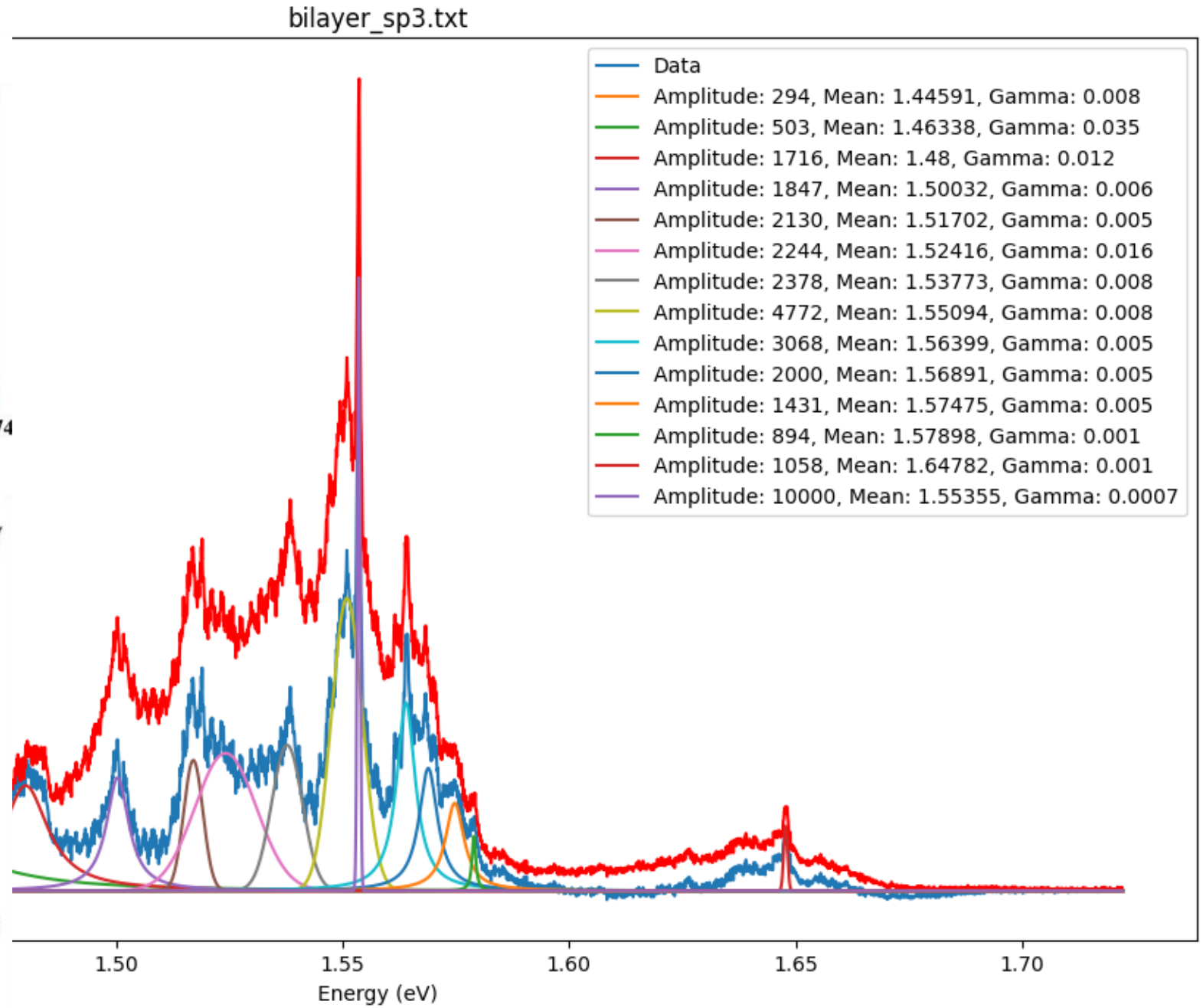


All spots

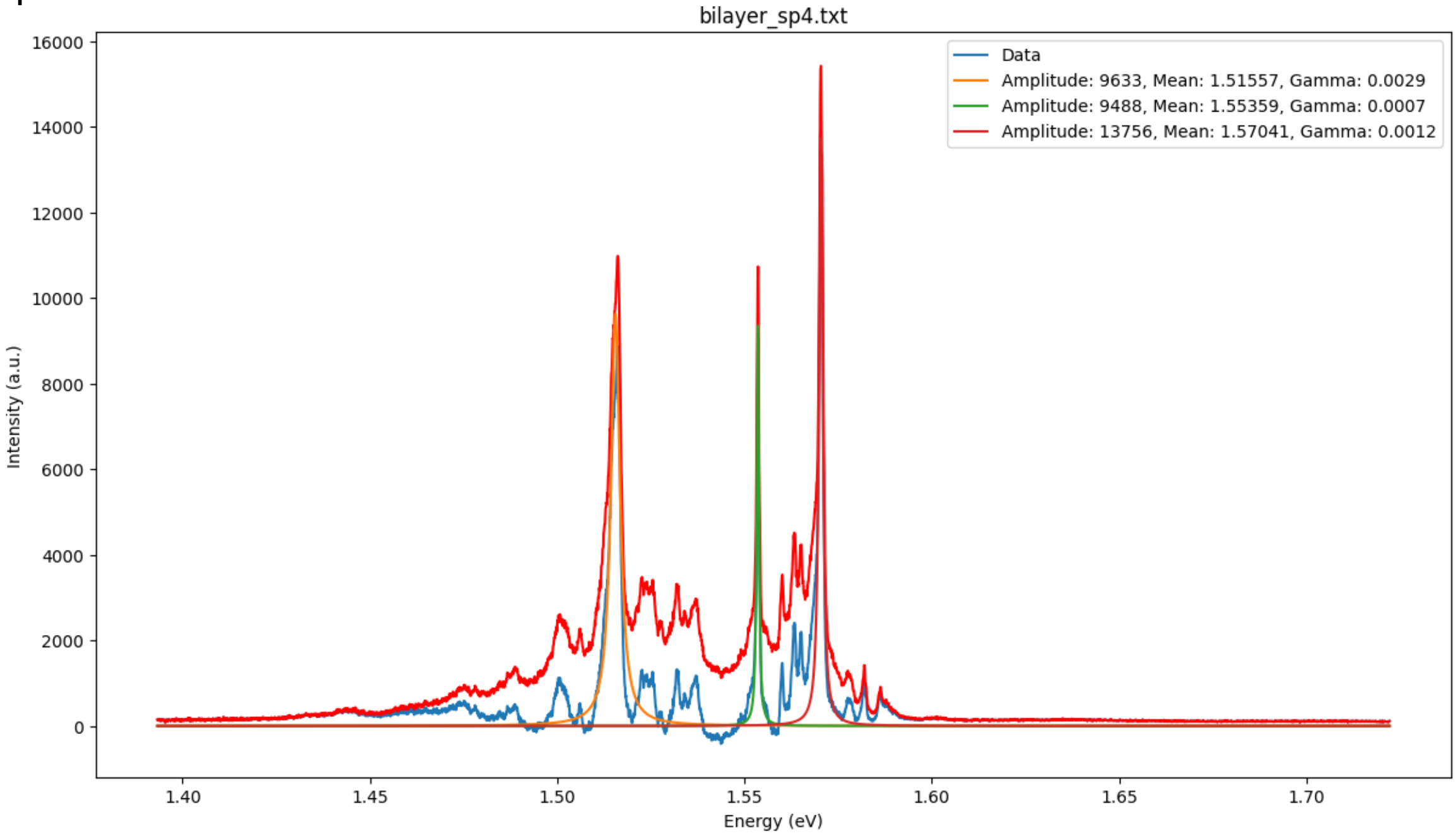

Data Plot
bilayer_sp1.txt
bilayer_sp2.txt
bilayer_sp3.txt
bilayer_sp4.txt
bilayer_sp5.txt
bilayer_sp6.txt
bilayer_sp7.txt
bilayer_sp8.txt
PL Intensity (a.u.)
10^4
10^3
10^2
10^1
1.40
1.45
1.50
1.55
1.60
1.65
Energy (eV)

Spot3

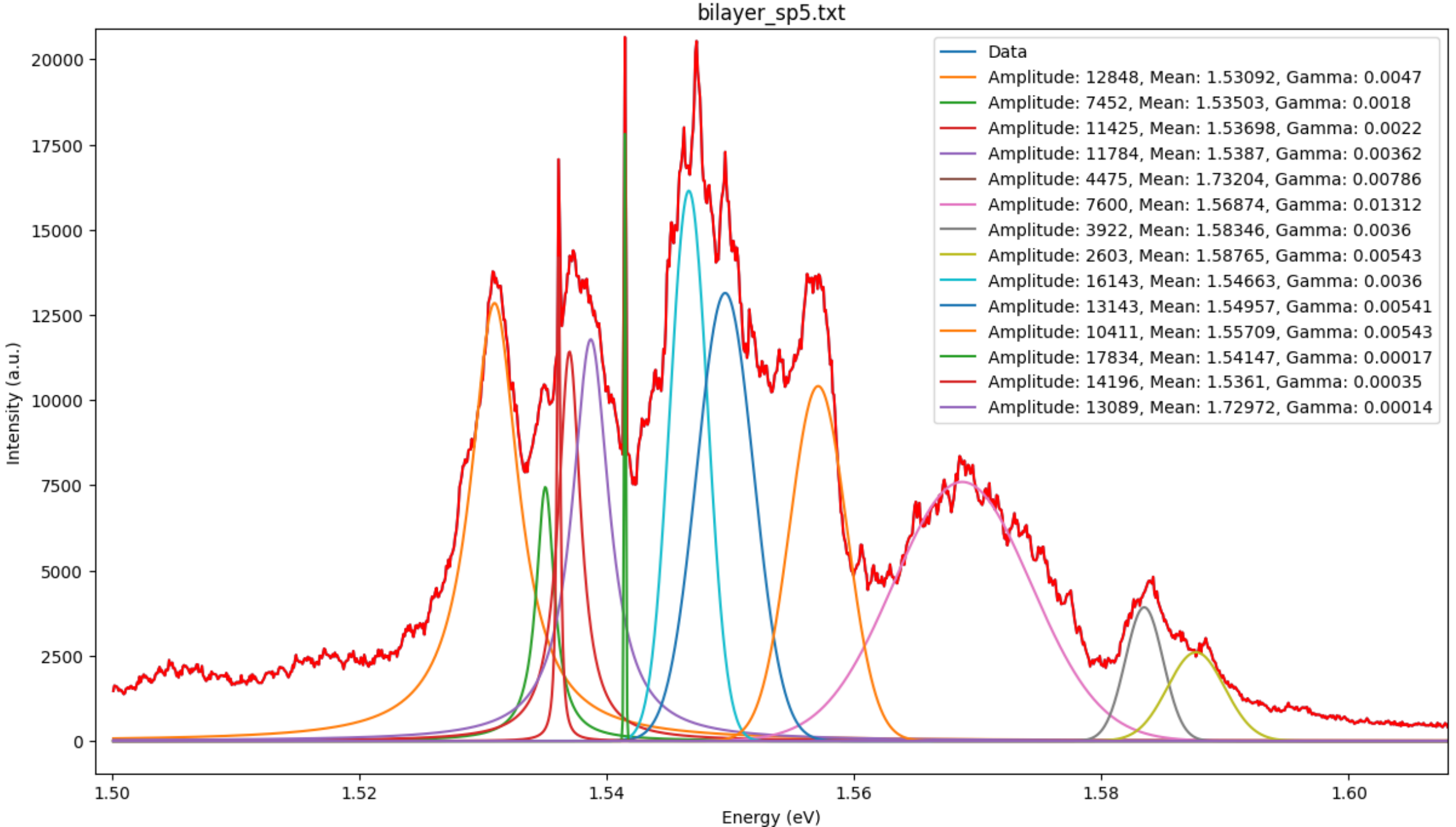

(a)
Intensity (a.u.)
X_0
1.42
1.50
1.58
1.66
1.74
Photon energy (eV)
(b)
Intensity
2400
36.00
E_z (V/nm)
0.2
0.1
0.0
-0.1
-0.2
X^2_{QK}
X^1_{QK}
X_{QK}
$X^3_{QΓ}$
$X^2_{QΓ}$
$X^1_{QΓ}$
$X_{QΓ}$
1.50
1.55
1.60
1.65
Photon energy (eV)


bilayer_sp3.txt
Data
Amplitude: 294, Mean: 1.44591, Gamma: 0.008
Amplitude: 503, Mean: 1.46338, Gamma: 0.035
Amplitude: 1716, Mean: 1.48, Gamma: 0.012
Amplitude: 1847, Mean: 1.50032, Gamma: 0.006
Amplitude: 2130, Mean: 1.51702, Gamma: 0.005
Amplitude: 2244, Mean: 1.52416, Gamma: 0.016
Amplitude: 2378, Mean: 1.53773, Gamma: 0.008
Amplitude: 4772, Mean: 1.55094, Gamma: 0.008
Amplitude: 3068, Mean: 1.56399, Gamma: 0.005
Amplitude: 2000, Mean: 1.56891, Gamma: 0.005
Amplitude: 1431, Mean: 1.57475, Gamma: 0.005
Amplitude: 894, Mean: 1.57898, Gamma: 0.001
Amplitude: 1058, Mean: 1.64782, Gamma: 0.001
Amplitude: 10000, Mean: 1.55355, Gamma: 0.0007
1.50
1.55
1.60
1.65
1.70
Energy (eV)

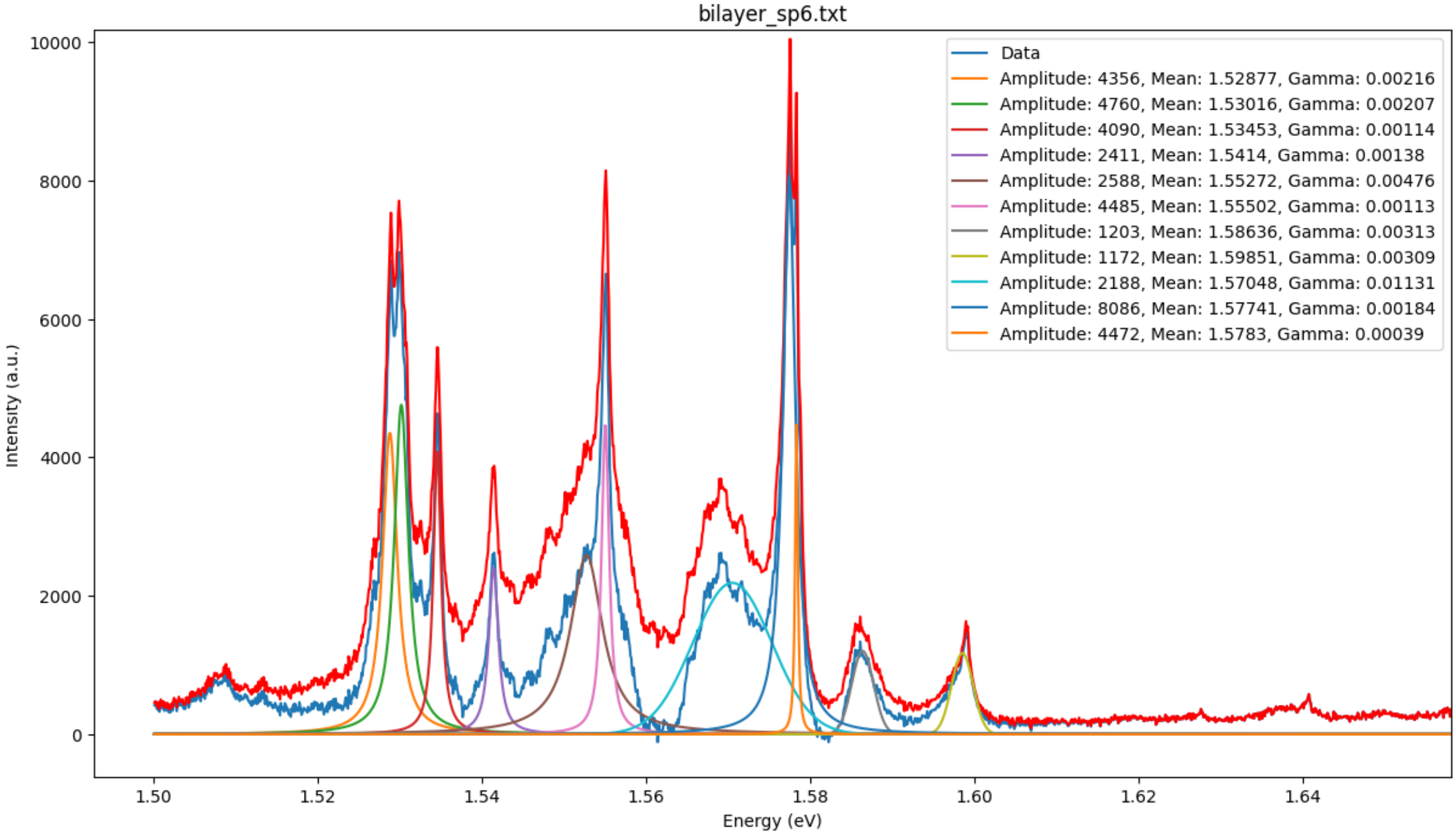
Spot4


bilayer_sp4.txt
Data
Amplitude: 9633, Mean: 1.51557, Gamma: 0.0029
Amplitude: 9488, Mean: 1.55359, Gamma: 0.0007
Amplitude: 13756, Mean: 1.57041, Gamma: 0.0012
Intensity (a.u.)
16000
14000
12000
10000
8000
6000
4000
2000
0
1.40
1.45
1.50
1.55
1.60
1.65
1.70
Energy (eV)

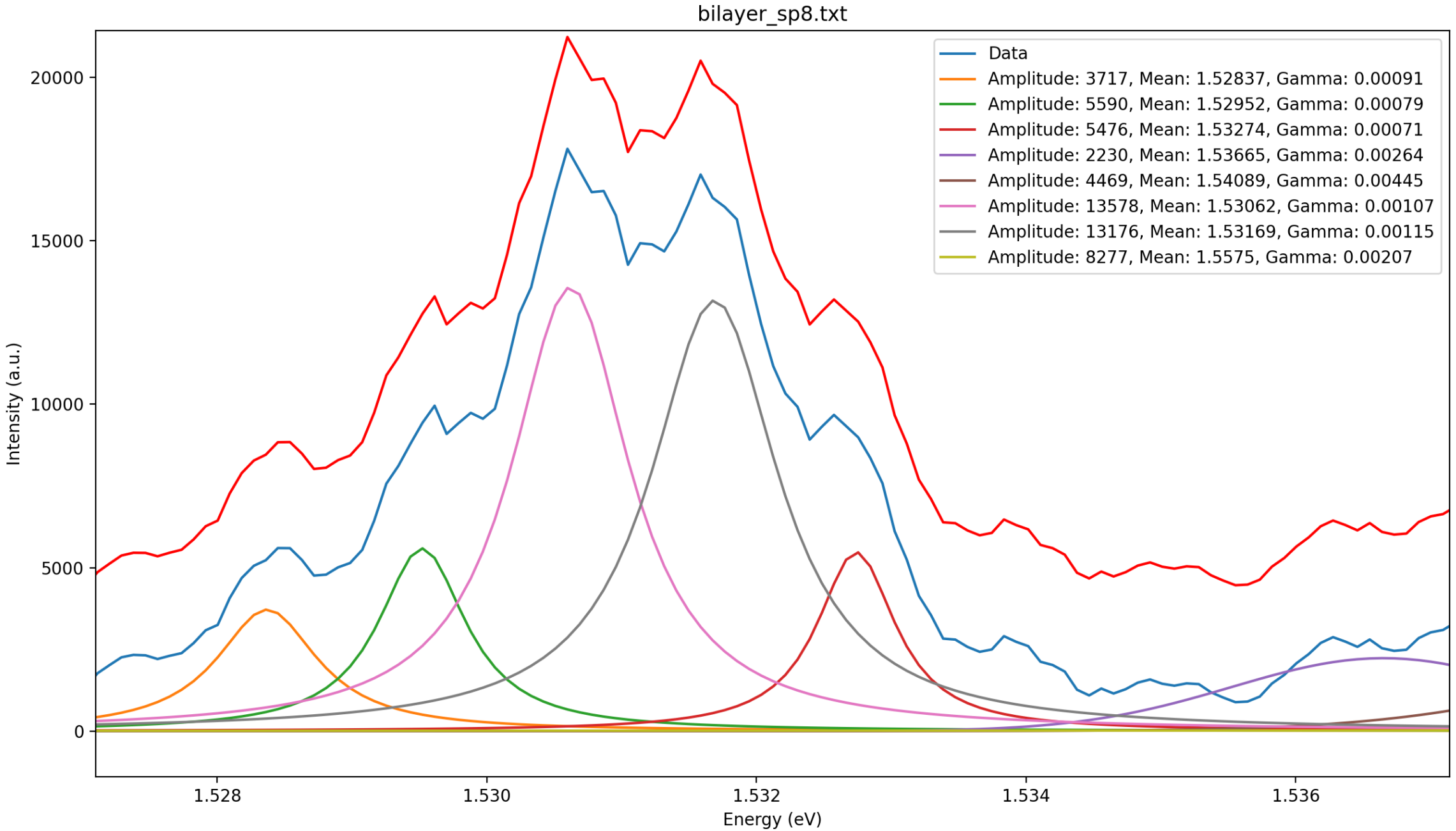
Spot5


bilayer_sp5.txt
Data
Amplitude: 12848, Mean: 1.53092, Gamma: 0.0047
Amplitude: 7452, Mean: 1.53503, Gamma: 0.0018
Amplitude: 11425, Mean: 1.53698, Gamma: 0.0022
Amplitude: 11784, Mean: 1.5387, Gamma: 0.00362
Amplitude: 4475, Mean: 1.73204, Gamma: 0.00786
Amplitude: 7600, Mean: 1.56874, Gamma: 0.01312
Amplitude: 3922, Mean: 1.58346, Gamma: 0.0036
Amplitude: 2603, Mean: 1.58765, Gamma: 0.00543
Amplitude: 16143, Mean: 1.54663, Gamma: 0.0036
Amplitude: 13143, Mean: 1.54957, Gamma: 0.00541
Amplitude: 10411, Mean: 1.55709, Gamma: 0.00543
Amplitude: 17834, Mean: 1.54147, Gamma: 0.00017
Amplitude: 14196, Mean: 1.5361, Gamma: 0.00035
Amplitude: 13089, Mean: 1.72972, Gamma: 0.00014
Intensity (a.u.)
0
2500
5000
7500
10000
12500
15000
17500
20000
1.50
1.52
1.54
1.56
1.58
1.60
Energy (eV)

Spot6


bilayer_sp6.txt
Data
Amplitude: 4356, Mean: 1.52877, Gamma: 0.00216
Amplitude: 4760, Mean: 1.53016, Gamma: 0.00207
Amplitude: 4090, Mean: 1.53453, Gamma: 0.00114
Amplitude: 2411, Mean: 1.5414, Gamma: 0.00138
Amplitude: 2588, Mean: 1.55272, Gamma: 0.00476
Amplitude: 4485, Mean: 1.55502, Gamma: 0.00113
Amplitude: 1203, Mean: 1.58636, Gamma: 0.00313
Amplitude: 1172, Mean: 1.59851, Gamma: 0.00309
Amplitude: 2188, Mean: 1.57048, Gamma: 0.01131
Amplitude: 8086, Mean: 1.57741, Gamma: 0.00184
Amplitude: 4472, Mean: 1.5783, Gamma: 0.00039
Intensity (a.u.)
10000
8000
6000
4000
2000
0
1.50
1.52
1.54
1.56
1.58
1.60
1.62
1.64
Energy (eV)

Spot8


bilayer_sp8.txt
Data
Amplitude: 3717, Mean: 1.52837, Gamma: 0.00091
Amplitude: 5590, Mean: 1.52952, Gamma: 0.00079
Amplitude: 5476, Mean: 1.53274, Gamma: 0.00071
Amplitude: 2230, Mean: 1.53665, Gamma: 0.00264
Amplitude: 4469, Mean: 1.54089, Gamma: 0.00445
Amplitude: 13578, Mean: 1.53062, Gamma: 0.00107
Amplitude: 13176, Mean: 1.53169, Gamma: 0.00115
Amplitude: 8277, Mean: 1.5575, Gamma: 0.00207
Intensity (a.u.)
20000
15000
10000
5000
0
1.528
1.530
1.532
1.534
1.536
Energy (eV)

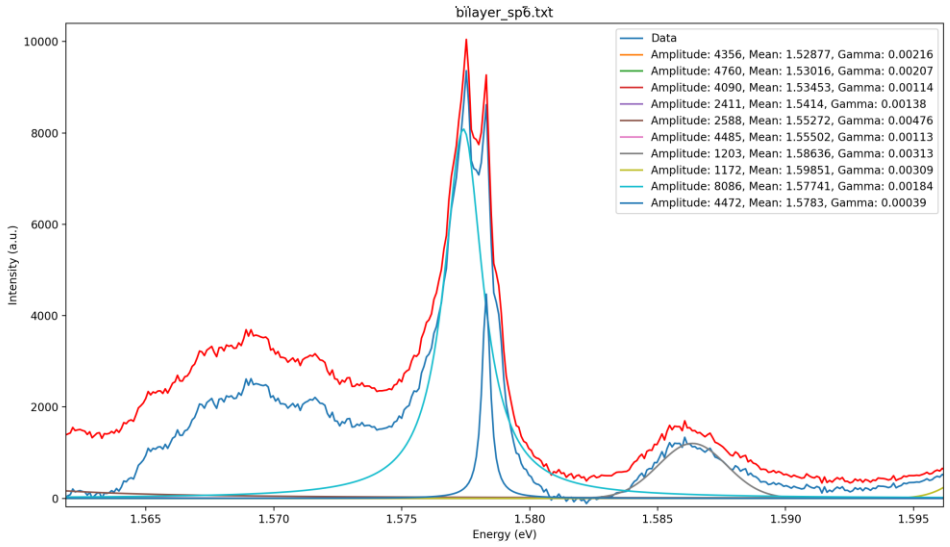
Analyze

Spot	FWHM/meV	Gaussian/meV	Lorentz/meV	Energy dif/meV	Lifetime/ps
Spot3	8.15	8.02	0.70	2.61	0.94
Spot5	6.27	5.41	3.60(0.14)	2.94	4.70
Spot6	5.31	4.76	1.13(0.39)	2.30	1.71
Spot8	4.35	3.80	2.07(0.71)	2.54	0.93

FWHM: Total emission linewidth

Lorentz: ZPL linewidth Gaussian: PSB linewidth
Energy difference: ZPL and PSB (Fig1)

Energy difference between main peaks?: 0.9meV (Fig2)



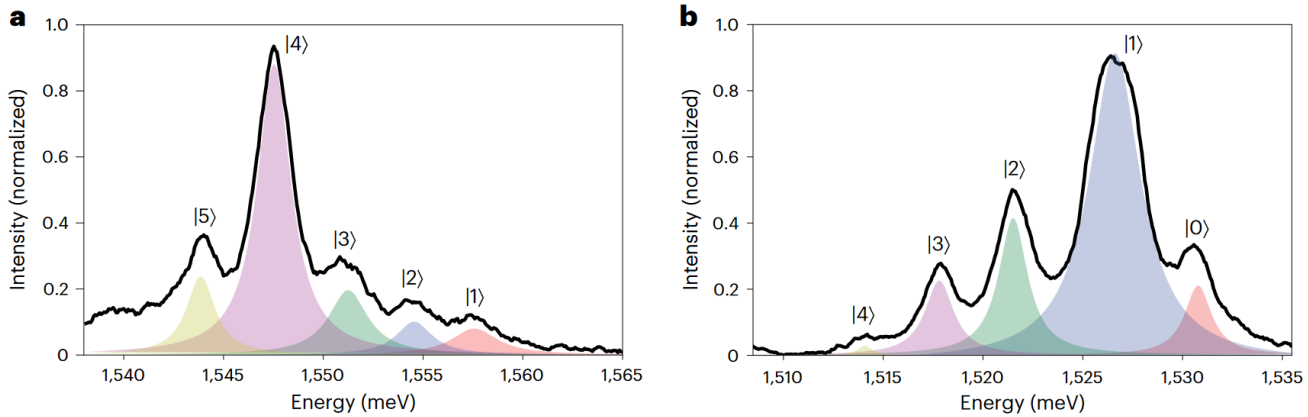
ively. In practice, spectroscopic linewidths are subject to various broadening mechanisms, thus limiting their utility for determination of dephasing time. Broadening mechanisms can include spectral diffusion, the dynamic nature of the local environment, exciton–phonon interaction, and the diffusive nature of the band-edge exciton.[8,38] For example, the band-

Less integration time--- suppress inhomogeneous contributions to PL

3 kinds of peaks and their side bands--- 2main(0.9meV), 1Gaussian bigger/smaller(2-3meV)

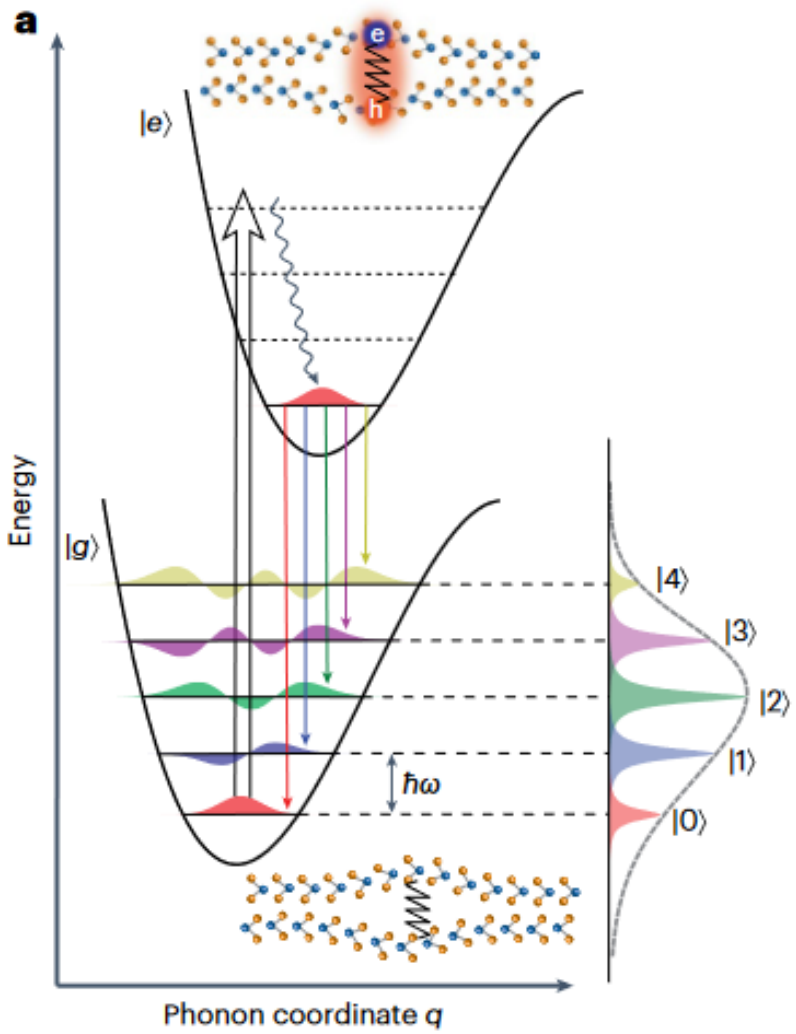
Momentum-indirect excitons? Defect-bound exciton peak? (for WSe2 3-5meV)

Need higher resolution of spectrometer?



https://doi.org/10.1039/C9NR02175B

Explanation



The ground state of the QE and its excited state when an IX is generated can be modelled with two potential energy surfaces.

At low temperatures, the QE at the ground-state PES has nearly zero phonon occupancy. When an exciton is created by the pump laser, the QE is excited into a higher-energy PES and quickly relaxes to its zero phonon level.

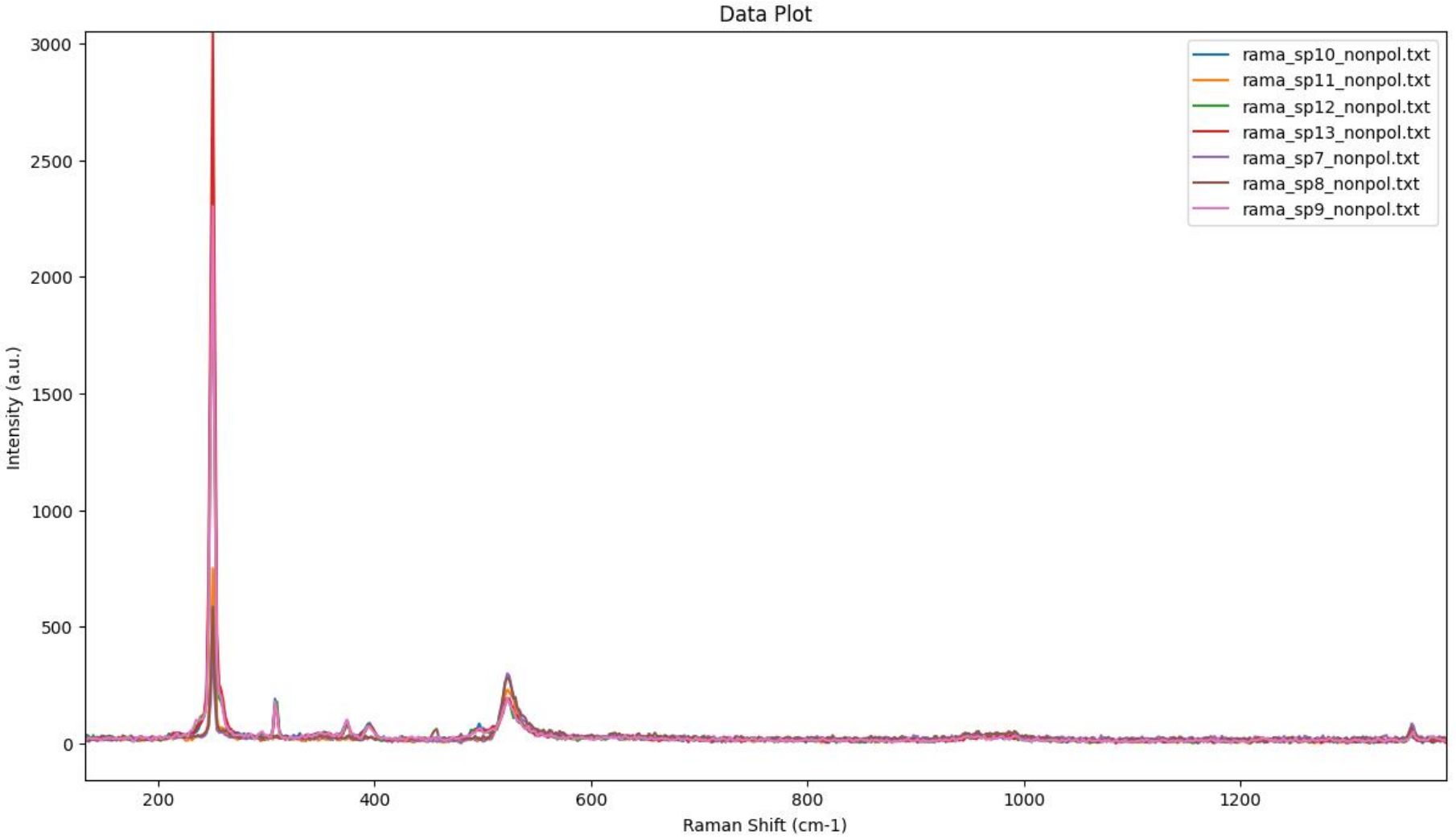
Upon exciton recombination, the QE emits a single photon and relaxes to its ground-state PES but at an elevated phonon state. As a result, the energy of the emitted photon is Stokes-shifted from the zero-phonon line, forming phonon replicas spaced by the phonon energy.

https://doi.org/10.1038/s41565-023-01410-6

Supplementary Table 1. Summary of IX quantum emitters that have been measured.

Emitter	Device	Energy tuning range (meV)	Min Energy (meV)	Max Energy (meV)	Dipole moment (e·nm)	Exciton Transition Types	Resolved Phonon Lines
QE1	A	6.35	1547.0	1553.4	0.341	KΓ	yes
QE3	A	11.5	1520.0	1531.5	0.649	KK	yes
QE4	A	6.26	1549.9	1556.2	0.377	KΓ	yes
QE5	A	9.75	1490.6	1500.4	0.668	KK	no
QE6	A	6.96	1522.2	1529.2	0.622	KK	yes
QE7	B	11.6	1543.8	1555.4	0.599	KK	yes
QE8	B	12.8	1494.2	1507.0	0.321	KΓ	no
QE9	B	11.6	1555.1	1566.7	0.331	KΓ	yes
QE10	B	6.20	1538.3	1544.5	0.586	KK	no
QE11	B	8.94	1595.8	1604.8	0.357	KΓ	no
QE12	B	5.20	1540.2	1545.4	0.315	KΓ	yes
QE13	E	7.90	1544.6	1552.5	0.459	QK	yes
QE14	E	8.67	1535.8	1544.5	0.480	QK	no
QE15	E	5.46	1525.4	1530.8	0.493	QK	yes
QE16	E	8.26	1545.8	1554.1	0.455	QK	yes
QE17	E	8.65	1520.6	1529.2	0.609	KK	yes
QE18	C	8.61	1494.1	1502.7	0.370	KΓ	no
QE19	C	20.2	1534.2	1554.5	0.657	KK	yes
QE20	D	8.43	1502.8	1511.3	0.471	QK	no

Raman

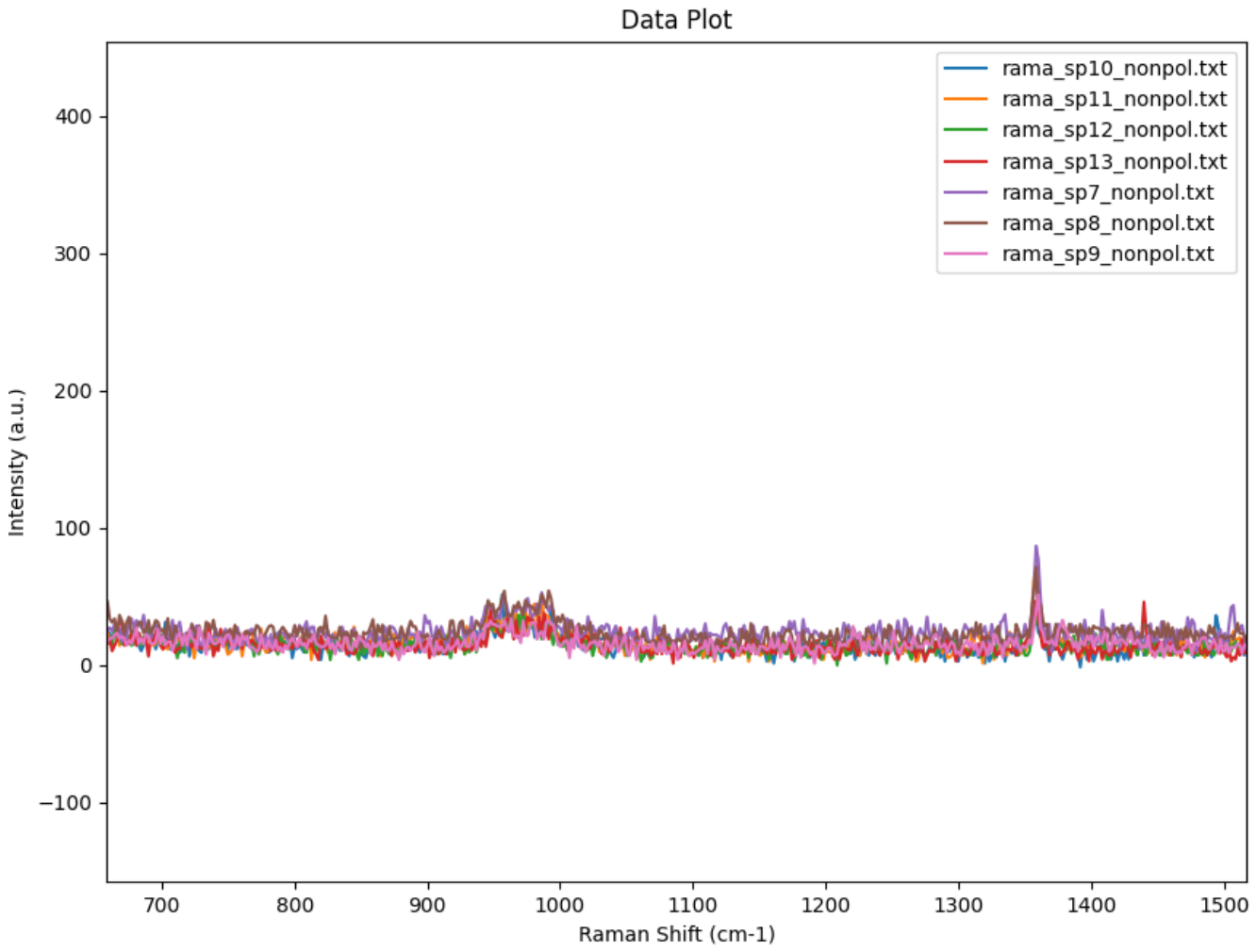


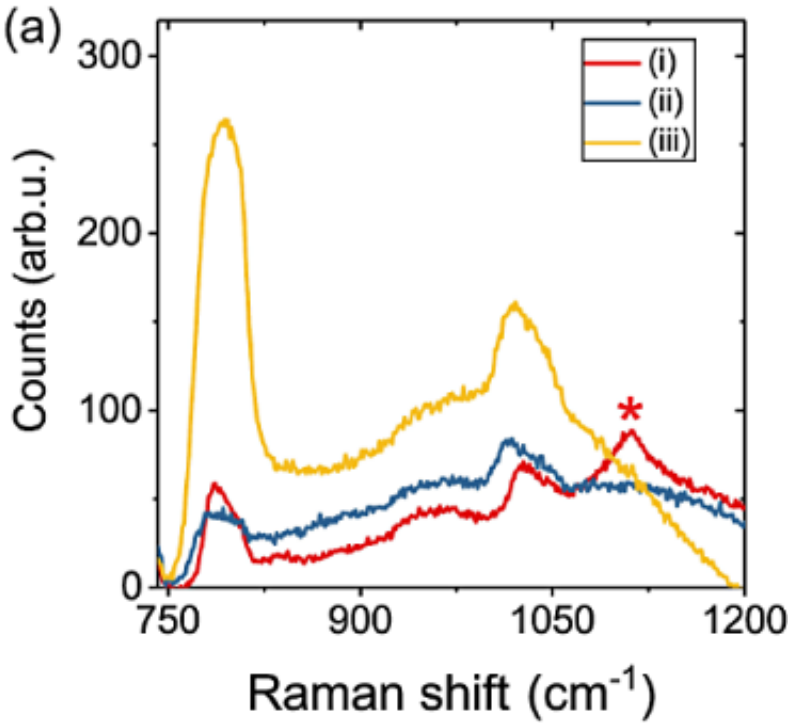
Main: E_{2g}^1/A_g Side band: $2LA(M)$

309cm^{-1} : B_{2g}^1 375cm^{-1} : $E_g + LA(M)$

395cm^{-1}: $3LA(M)$ 520cm^{-1} : Si substrate 1370cm^{-1} : hBN E_{2g}

$960cm^{-1}$ peak?

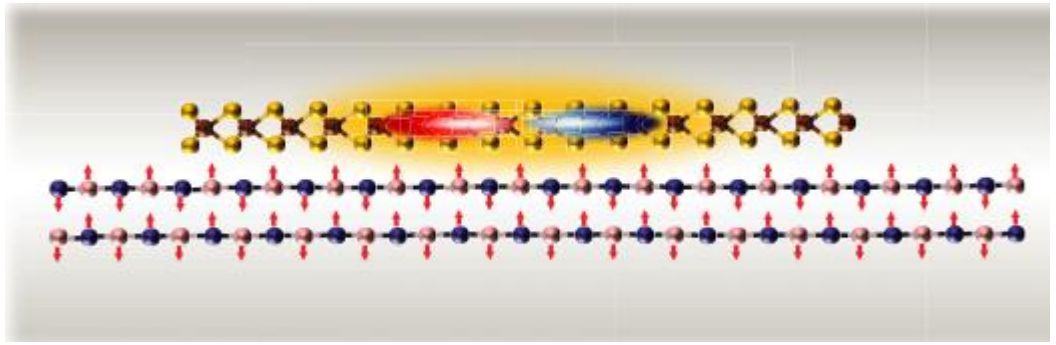


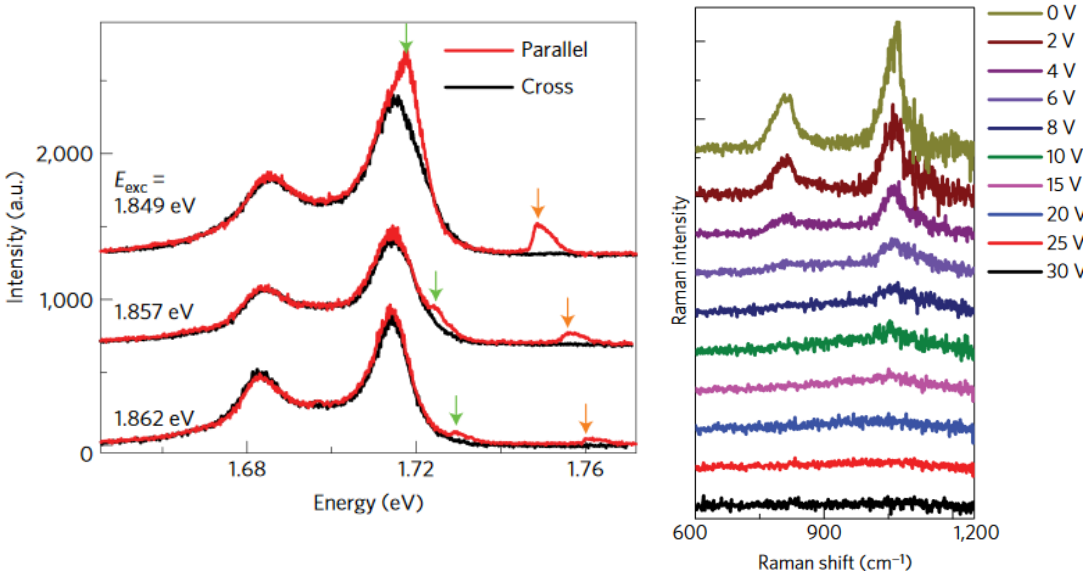


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Interlayer electron–phonon coupling in WSe_2/hBN heterostructures

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The vibration modes of the hBN lattice can modulate the behavior of electrons and holes in WSe2 thus enabling interlayer electron phonon coupling.

New Raman modes from hBN ZO phonon and the combinatorial hBN ZO and WSe2 A1g phonon in WSe2/hBN heterostructures.

Induce Polarization/ Gate voltage as parameters?
Scan <100cm^{-1} Raman for BM phonons?